

FINAL YEAR PROJECT (FYP)

Literature Review

“A Smart Learning and Attendance Management System”

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1. Introduction

Futsal is becoming popular but the management is predominantly manual and structured, using phone calls, social media, or paper books. Players have many problems related booking courts, futsal owners have problem with scheduling and cancellations, and event preparation.

This project is a MERN-stack based futsal booking management tool enable players to make a futsal booking within minutes, owners to control slots and revenue through analytics. Compared to the current systems, PlayPal will be characterized by better functionality, including real-time availability, team matching, split-payments, ratings/endorsements, injury-tracking, notifications, highlights and tournament automations. This makes the project a booking tool but an entire sports ecosystem to players and owners together under one digital platform.

2. Aims and Objectives

Aim:

To create and build a futsal management system which simplifies the booking process, enhances the functionality of the owners contributes to youth sport involvement and allows the business to be up-scaled in the future.

Objectives:

- Create an easy access site where players can search and reserve futsal courts at any given time
- Develop features like team match making, endorsements and reviews which are player-community features.
- Provide the dashboards to futsal owners to handle bookings, cancellations, prices and income.
- Combine secure online payment systems to create trust and transparency.
- Add tournaments and sports events to connect the players
- Send notifications, reminders and calendar synchronization to players
- Introduce gamification (leaderboards, achievements) to improve the interest of players.
- Injury/health monitoring
- Study scalability options to other sport and recreation facilities in the future.

3. Problem Statement

i. Player-Specific Problems

- Difficulty locating local available futsal courts in real time
- Encounters problems with building entire teams or locating opponents.
- No reliable information available regarding futsal quality, reliability of players and slot availability
- Unknown about the upcoming events and tournaments
- No system to monitor performance, achievements, or progress
- Feel discouraged to play on the regular basis because of the lack of gamification or rewards.

ii. Owner-Specific Problems

- Loss of revenue due to last minute cancellation and no shows
- Relies on cash-only, which causes conflicts and incompatibility
- Absence of proper dashboards and analytics to track peak hours, customer loyalty and revenue trends
- Difficulty in scheduling and control of slots manually

iii. Community-Specific Problems

- Poor player involvement since there are no digital platform such as matchmaking and endorsements
- Lack of performance monitoring to promote healthy competition

4. Project as a Solution

PlayPal has the solutions to these problems as it offers:

- Live booking and futsal availability
- Match making and finding opponents on a team basis
- Split payment
- Automation of tournaments and event timetables
- Reviews, ratings and
- Safe play injury tracking
- Push notification and calendar sync

5. Scope and Limitations

Scope:

- Integrate web access to players, futsal owners and admins.
- Allow futsal to be booked in real-time, and availability checked with minimal confusion and overlapping bookings.
- Push notification delivery and calendar sync of booking reminders, matches ahead
- Allow team matchmaking, endorsements, analytics, tournaments and event management
- Give futsal owners the ability to control slots, price, block unavailable times and see revenue/attendance statistics.

Limitations:

- Digital payments depend on API third-party (eSewa, Khalti, Stripe); third-party service providers might crash or degrade.
- Scheduling: Automated fixtures and availability requires correct initial data by futsal owners/admins: system output can be impacted by human mistakes
- Facility is fitted to futsal; other sports (badminton, cricket, gyms) will be added later in the stages.
- At present, web-based, mobile application to be created later to allow more people to access it.

6. Functional Decomposition Diagram (FDD)

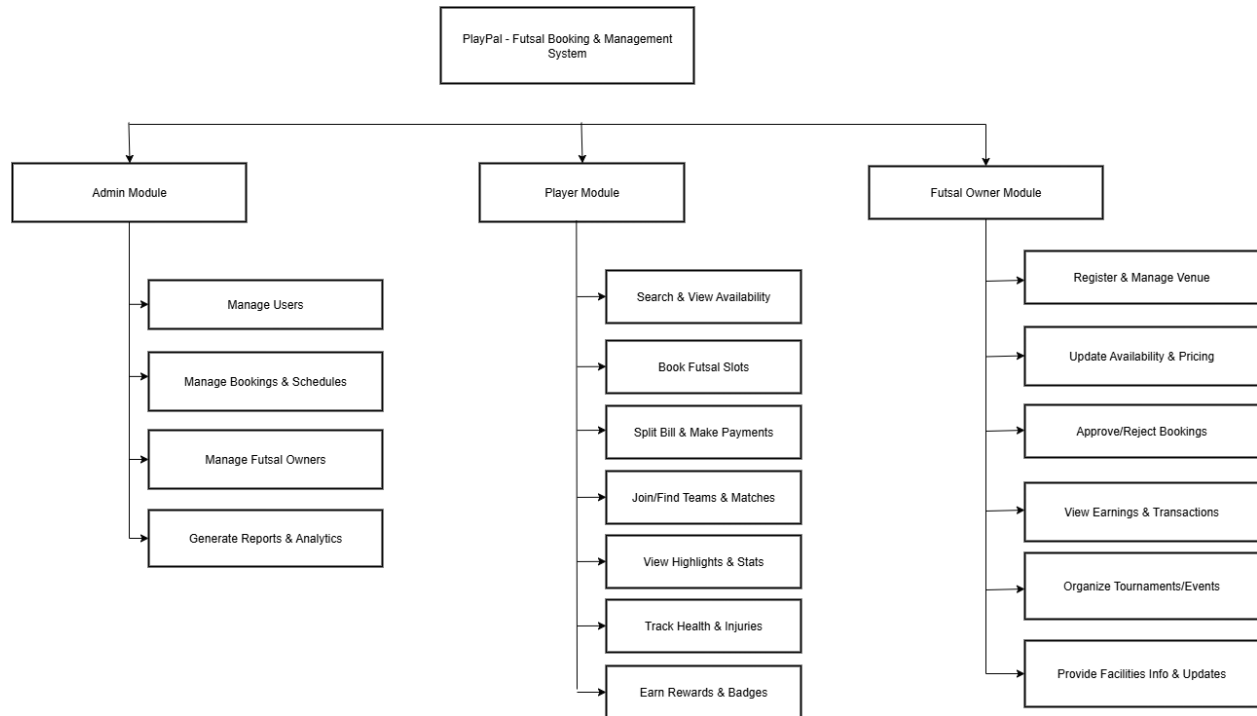


Figure 1:FDD Diagram

7. Artifacts (Proposed)

The majority of the available resources such as Playo, JustPlay, and FutsalHub only aim at the single booking or player search. Other key areas like payment management, management of the tournaments, and tracking of injuries are ignored. This leads to a disjointed experience in which players, owners must depend on various systems or offline communication to satisfy their needs.

The artefact proposed, PlayPal, tries to address this issue by putting it all together in a single platform. Not just a booking but a direct solution to concrete issues, such as double-bookings, cash issues, team building issues, parental involvement, injury-prevention, and even gamification-based motivation. This is tied directly to the academic question: How can a digital platform based on MERN enhance efficiency, transparency, and engagement in the futsal booking and management process, and help families and kids use sports? Combining booking, payments, analytics and community into a single ecosystem, PlayPal can enhance the experience of players, provide better management to the owners of futsal facilities.

Subsystems:

i. Booking and Availability Subsystem:

- Input: Request by players, futsal schedules
- Process: slot check, conflict inspection, approval of booking (in real-time)
- Output: updated booking calendar, notifications

ii. Payment and Split Billing Subsystem:

- Input: player/team booking charges, method of payment (eSewa, Khalti, Stripe)
- Process: Secure transaction, auto-divide between players, refunding
- Output: Open payments, payment history

iii. Player Engagement and Matchmaking Subsystem:

- Input: Player profile, team requests, endorsements
- Output: Teambuilding, opponent suggestion, endorsements and ratings
- Output: Balanced teams, reliable matchups

v. Tournament and Event Management Subsystem:

- Input: Registration of teams, tournament information
- Process: Auto – fixture, schedule updates
- Output: Tournament brackets, event calendars, simplified coordination.

vi. Highlights Subsystem:

- Input: Recordings of the match
- Process: Production, storage of clips, streaming to players.
- Output: On-demand highlights, shared match content.

vii. Health and Injury Tracking Subsystem:

- Input: Status of player injury, health
- Process: Injury notification, prevention notification, ineligibility to matchmaking due to unsafe.
- Output: Safer play, prevention of aggravated injuries.

8. Initial Research (Literature Review)

Paper Review

i. Digital transformation in the sports industry

According to the latest literature, the digital transformation can transform the management of sports facilities, make them more efficient, provide improved working processes, and increased user experience. Internet resources can actually be useful to increase the utilization rates, volume of work of the administration and the satisfaction of the customer in the case the services of the booking serve appear to be more efficient and safe, as the outcomes of the research works present (Romano, 2020). Equally,

research highlights how technologies like wearable devices, artificial intelligence-driven analytics, virtual reality, and digital health devices can enhance the performance of athletes and audiences as well as support marketing, sponsorship, and operations-related decision-making (Acar, 2025).

On the basis of these results, scholars in the sport management environment propose slow, tactical digital transformation, that is to say that companies should start by taking minor actions and develop (Deloitte, 2023). (Romano, 2020)also highlights the prospects of applying data analytics, IoT, and AI to resource management, injury prevention, and decision-making, which is also the direct route to the proposed PlayPal architecture. Collectively, these articles underline the importance of including such capabilities as real-time availability, analytics dashboard, gamification, and health tracking as a way of making the operations more efficient, yet they also generate interest, which is a definite confirmation of the idea of PlayPal as an all-inclusive digital futsal ecosystem.

ii. Gamify, engage, build loyalty: Exploring the benefits of gameful experiences in branded sports apps

The incorporation of gamified components into branded sports apps contributes greatly to user motivation and their loyalty. The researchers find that gameful experiences, which are promoted by game mechanics like badges, leaderboards, and progress tracking, promote closer emotional involvement and a desire to keep using the application. This will create a higher retention and loyalty as they will be more attached to the brand therefore reducing the retention rate (Jorge and Palau-Saumell, et al., 2024).

Moreover, the study reveals that customer brand engagement (CBE) has a major mediating effect between gamified experiences and user loyalty. It takes a positive, emotionally resonant experience to make users of the app cognitively and affectively attached to the brand, thus more willing to remain engaged and less prone to change platforms. Simply put, a highly crafted gameful experience leads to increased brand engagement, subsequently to loyalty and continued usage, which are essential findings helpful in justifying the importance of gamification in the design of PlayPal .

iii. Exploring the Impact of Gamification on Users' Engagement for Sustainable Development: A Case Study in Brand Applications

The research questions how brand loyalty and user engagement can be achieved through the use of game mechanics in brand applications (self-achievement and team-achievement features). Based on the MDA (Mechanics Dynamics Aesthetics) framework, it shows that such mechanics as challenges with self and

team can enhance the positive emotions, interest, and brand stickiness of users, as the most effective way to gamify is to balance fun and rewarding experience with feedback and incentives (Lu, et al., 2020).

Collaborative and team-based challenges and game mechanics promote interaction, cooperation, and competition between users and this reinforces the feeling of community and long-term loyalty. In the case of PayPal, it implies the implementation of gamification not only as an add-on aimed at making the service enjoyable, but as one that is carefully planned and executed to ensure that users remain engaged, motivated, and loyal to the platform.

iv. Integrated QR payments system (QRIS): cashless payment solution in developing country from merchant perspective

The research enabled (Rafferty, et al., 2022) to examine the QR-based framework of adoption of cashless payment system by small enterprises in developing countries. They found that lack of digital infrastructure, low digital literacy and over-reliance on cash are some of the challenges that merchants encounter. The digital payments are more likely to be adopted by merchants, however, when they can trust them, find it convenient, and see the benefits of using them explicitly. It shows that even in an environment, where cash is a primary concern, a possibility to improve business operations and customer experience by implementing safe and convenient digital payments exists.

In the case of the PayPal Futsal Booking System, a QR or cashless payment system can facilitate the booking process, minimize the problems associated with cash transactions, and make the system more user-friendly. The same way a small trader can enjoy the power of digital payment, PayPal users (players and futsal organizers) can make their bookings in seconds, monitor their transactions, and have a smooth experience. The system needs to be easy and safe to ensure that people adopt and use the platform.

V. Resource scheduling in edge computing for booking system

Recent surveys have shown that resource scheduling would help to multiply by far the potentials to provide digital services (real-time responsiveness, low latency and workload optimization) in edge computing (Luo, et al., 2021). As far as booking and management systems are concerned, these findings can be translated as the opportunity to serve multiple requests simultaneously without any conflicts, the equitable allocation of time slots, and the system stability even during the busiest period. In the study, proper scheduling policies like load balancing, priority-based allocation, provision of resources dynamically and adaptively to minimize delays and elimination of problems like double booking that is highly sensitive to applications such as PayPal whereby the user must be fully aware of

The researchers also note that efficiency is not the sole factor to consider when designing scheduling frameworks, and additional user-friendly principles, such as fairness, transparency, and scalability, must be considered (Luo, et al., 2021). This applies directly to PlayPal because futsal facilities need to manage slots in real-time, update them dynamically when things go wrong, and provide notifications to users automatically. Applying these principles in the context of scheduling will assist in ensuring the positive user experience, developing trust in the platform, and reducing the administrative burden of facility managers. Taken together, this literature indicates the need to make edge-conscious scheduling, adaptive allocation, and conflict-prevention contrivances part of the PlayPal architecture to justify its worth as a trustworthy and future-proof futsal booking ecosystem the suitably placed slot.

System Review

i. UrbanPro Sports Class:

UrbanPro Sports Classes is mainly an intermediary between children and parents and sports training programs. It is a skill development organization that provides formal coaching lessons in various sports. It is strong in educational departments of sport particularly among the young learners and discovery are the two essential qualities about it in classes and training classes.

However, UrbanPro does not provide live futsal booking, and futsal-specific management services, which limit its use to players or teams that need to play on the spot. PlayPal fills this gap by incorporating real-time futsal booking, instant slot visibility and parental management capabilities, so as to combine the strong UrbanPro youth engagement with the immediate practicality that comes with futsal play. This forms a complete ecosystem to support training and gameplay.

ii. JustPlay:

JustPlay aims at making things easier when it comes to finding a teammate and creating a group to play a casual game. It provides a matchmaking service that brings together individuals who would like to play without a team, thereby building a social game based on impromptu play. It revolves around social discovery and uniting people via sports.

Despite its team-finding advantage, JustPlay does not integrate futsal-specific bookings or venue management. This means users can form groups but struggle to secure a slot or manage a futsal session seamlessly. PlayPal resolves this by combining real-time futsal booking with team-matching tools, ensuring that players not only find teammates but also instantly secure courts, streamlining the entire experience from group creation to gameplay.

iii. BookMyPlay:

BookMyPlay is a multi-sport booking platform that deals with badminton, tennis, football, cricket, and many others. It possesses an enormous degree of accessibility which is found in locating venues, booking slots as well as has installed payments that are accessible to different kinds of sports. The site is strong due to its variety because it is easy to people who participate in different sports.

The disadvantage of this breadth is that it lacks depth specific to futsal and lacks loyalty and community aspects. Consequently, no special attention is paid to futsal players, and user retention is lower in the absence of rewards and long-term engagement tools. PlayPal addresses this directly through the creation of a futsal-only ecosystem that incorporates loyalty programs, community-building features, and repeat-player incentives to make futsal players feel valued and benefit when they play regularly.

9. Methodology

Development Approach: Agile (Scrum)

This project has plenty of mini-applications as well user management, scheduling, payment, owner dashboard, booking, etc. It could be a good idea to divide it into several chunks and work on. Hence Agile

simplify this application to develop, test and enhance at every step as per feedback. For the testing, unit testing, Integration testing, user acceptance testing to prove that it works, it is easy to use and it works well.

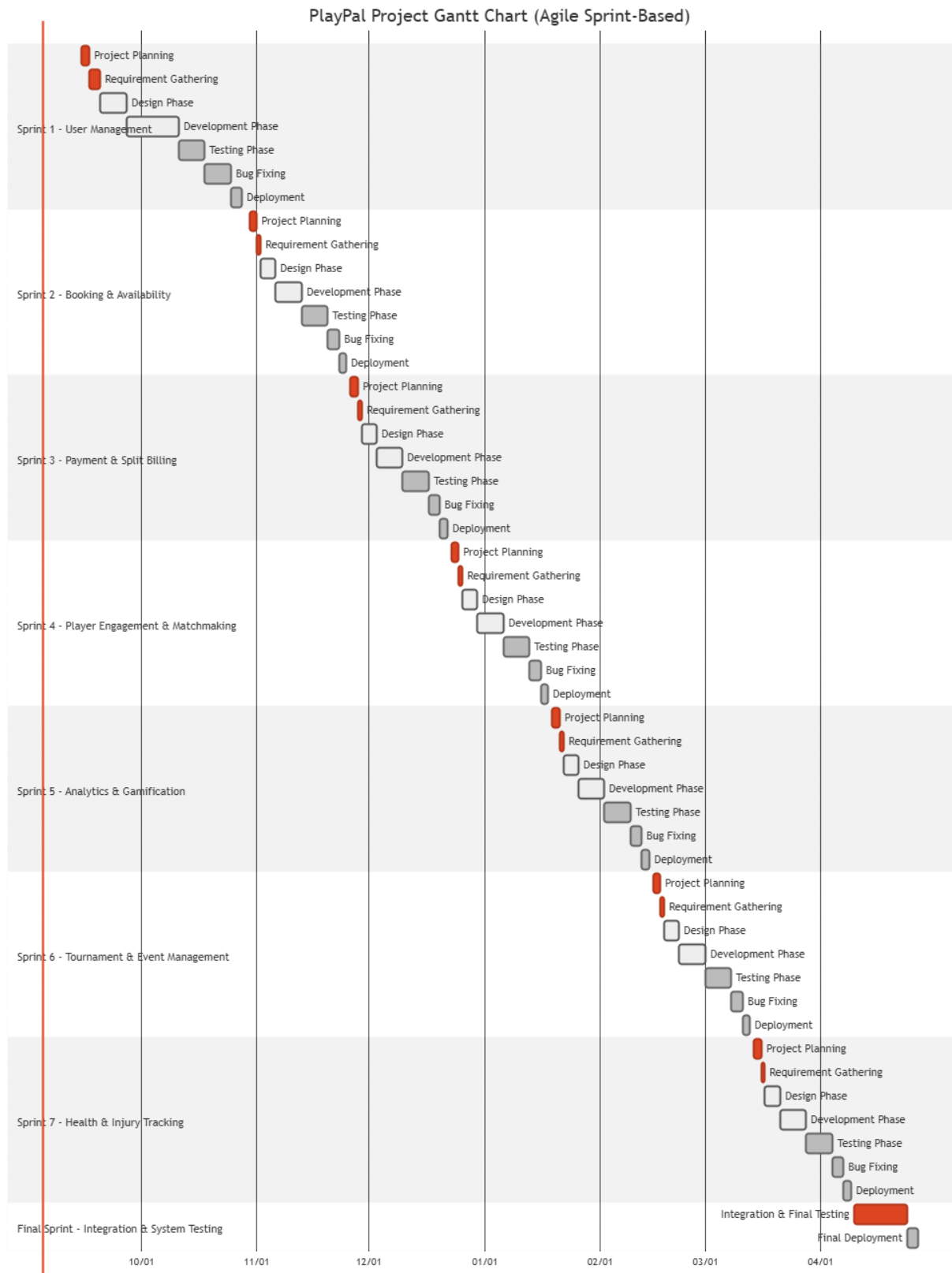


Figure 2: Gantt Chart

10. Tools and Technologies

- Frontend: React.js
- Backend: Node.js with Express.js
- Database: MongoDB (NoSQL, easily adaptable for the attendance, schedules)
- Payments: Stripe API, eSewa, Khalti
- Version Control: GitHub
- IDE: VS Code
- Design: Figma
- Deployment: Cloud (Heroku/AWS)

11. Summary

The project PlayPal - Futsal Booking and Management System aims to address the real issues encountered in the running of futsal today. The system provides convenience to the players, professionalism to the futsal owners, and active involvement players by providing services like real-time availability, team matchmaking, secure split payments, endorsements, analytics, live streaming. PlayPal is also a full ecosystem of sports, not a reservation tool as the current solutions do because they only ensure basic booking and nothing more.

Academically, this project shows how successfully the MERN stack was used to create a scalable web application integrating booking, payments, analytics, and communication on a single platform. On the business side, it provides an opportunity to sustainable growth via revenue models such as commission per booking, future visibility of futsal owners, advertisement. PlayPal has a potential to be taken as a useful academic contribution, as well as a viable solution with a commercial potential with its innovative design, solid research basis and it is clear that it can be scaled.

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